

Jiyanshu Singh

B.Tech Biotechnology student at NIT Rourkela | Building ML systems

Contact

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Education

B.Tech Biotechnology · National Institute of Technology, Rourkela

Expected 2027 · CGPA: 7.09/10 · Relevant coursework: Data Structures & Algorithms, Statistics, Mathematics for Engineers

Experience

AI Intern · Digitaaztrans Technologies (Remote) Jan 2025 – Present

MLEngineering: Designed, deployed, and operated an ML-filtered multi-strategy trading engine on Google Cloud Platform serving live market data.

Deployment: Containerized the system with Docker; orchestrated 6 parallel strategies on GCP with PostgreSQL logging of 38,500+ labeled trades.

Validation: Applied walk-forward testing across 3 disjoint bull/bear market windows to ensure out-of-sample robustness.

Performance: The ML trade filter improved simulated portfolio from a -■225,000 drawdown to +■93,000 profit.

Freelance Data Scientist / ML Engineer Dec 2023 – Present

End-to-end delivery: Built and deployed 4 production web applications (A/B Testing Dashboard, Sales CRM, Invoice Generator, Hydrogel Portal) — all with Dockerized FastAPI/Flask backends on Render/GitHub.

Statistical consulting: Delivered A/B test analysis (p-values, Bayesian Monte Carlo, power analysis) for small clients.

Projects

AI Trading Engine [Code]

Problem: A naive strategy lost ■225k; an XGBoost ML filter that labels 38,500+ trades as net-profitable-after-costs turned it into +■93k profit.

Key results (OOS-validated): 95% reduction in false signals; walk-forward validated across 3 market regimes (bear → consolidation → rally).

Tech stack: Python, XGBoost, FastAPI, Docker, GCP (Compute Engine + Cloud SQL), PostgreSQL, walk-forward backtesting framework.

A/B Testing Dashboard [Code]

Purpose: Multi-page statistical analysis app that analyzes experiment data and recommends Launch / Iterate / Kill using p-values, Bayesian Monte Carlo, power analysis, and sample size calculations.

Impact: Generates consulting-grade PDF reports for stakeholders — deployed at <https://ab-testing-dashboard.onrender.com>.

Tech stack: FastAPI, Bootstrap 5, SciPy, Docker, Render.

Skills

Languages: Python (expert), SQL, JavaScript

ML/AI: XGBoost, scikit-learn, walk-forward validation, Bayesian inference, statistical testing (SciPy), hypothesis testing, power analysis

Data: pandas, NumPy, PostgreSQL, FastAPI, Docker, GCP, REST APIs, Git

Achievements

- Deployed 5 production web applications (FastAPI/Flask + Docker + Render/GCP).
 - Labeled 38,500+ trades for ML filter; 95% false-signal reduction; walk-forward OOS validated across 3 market windows.
 - 350-sample peptide benchmark with leakage prevention; amyloid ROC-AUC 0.888.
- Skills I'm learning:** XGBoost, walk-forward testing, Bayesian A/B testing, biotech ML.